Ari Gestetner

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Education

Monash University | *Melbourne, Victoria, Australia*

Bachelor Commerce and Science in Mathematical Foundations of Econometrics, Mathematics and Computational Science **2023-2025**

- Awarded with summer and winter vacation research scholarships
- First place in an IMC algorithmic trading competition for Monash students hosted by Monash Association of Coding (MAC)

Experience

Teaching Associate **Feb 2024 - Present**

Monash University

- Achieved 89% student satisfaction rating exceeding the faculty average of 82% by providing engaging, individualised and interactive classes and explanations
- Collaborated with fellow TAs to deliver applied classes, sharing feedback to drive continuous teaching improvements and enhance student learning outcomes

Research Scholarships

Monash University

Nov 2024 - Feb 2025 - Dr. Buser Say *Python, SCIP, Constraint Programming*

- Continued extending SCIPPlan to allow users to provide a system of odes to be solved and used in the planning process supporting engineers and physicists in their work
- Incorporated function interpolation, introducing flexibility to the planning process and allowing for more complex problems to be solved

Jul 2024 - Dr. Buser Say *Python, SCIP, Constraint Programming*

- Developed SCIPPlan+, an automated planner for discrete and continuous time planning problems with stochastic domains
- Researched the applications and performance of SCIPPlan+ over a variety of domains

Nov 2023 - Feb 2024 - Dr. Buser Say *Python, SCIP, Constraint Programming*

- Refactored and extended SCIPPlan an automated planner for continuous time sequential planning problems over non-linear domains
- Assessed the performance of SCIPPlan in identifying optimal plans by formulating diverse problems with non-linear constraints
- Successfully derived feasible optimal solutions for a novel problem within a reasonable time frame

Jul 2023 - Dr. Buser Say *Python, Cplex, Gurobi, Seaborn, MILP*

- Investigated various linear programming constraint formulations based on neural network transition functions
- Benchmarked and presented different solvers and formulations finding results of over 2x decrease in solve time

Projects

Pandemic Prediction & Planning *Python, SciPy, Matplotlib*

- Explored complex disease spread dynamics by extending the standard SIR model to include real-world factors like birth and death rates to answer specific modelling questions.
- Utilised stochastic systems to create prediction intervals for possible outcomes and applied genetic algorithms to identify optimal lockdown schedules that minimise duration while keeping infections below a safe cap.

Retail Sales Forecasting *R, Fable, Tidyverse*

- Analysed multi-decade Queensland retail turnover data to identify long-term growth trends and predictable seasonal shopping patterns.
- Evaluated multiple statistical models to forecast future sales, specifically measuring the impact of major market disruptions like the 2020 pandemic and the speed of subsequent recoveries.

Housing Market Analysis *R*

- Investigated the relationship between house and apartment prices across Melbourne and Sydney to determine if different city markets move together in the long run.
- Developed models to perform impact analysis, measuring how local property prices react over time to unexpected market shocks or specific policy changes like urban rezoning.

Finance API *Python, FastAPI, BeautifulSoup, PyTest*

- Architected a high-speed tool for stock data retrieval, using multithreading to scrape financial statements from multiple sources simultaneously for a 5x increase in speed.
- Engineered a Google Sheets integration used to manage and track $15,000 in real-world investments, ensuring system stability through automated unit testing.

Technical Skills

Languages: Python, R, Typescript, C, C++17, SQL, HTML/CSS

Frameworks: Flask, FastAPI, PyTest, Vue3, Nuxt3, JUCE

Developer Tools: Git, Docker, RStudio, Linux, Vercel, GCP

Libraries: BeautifulSoup, Pandas, Matplotlib, Numpy/Scipy, Pytorch, Tidyverse, Fable, Cplex, Gurobi, SCIP

Academic Concepts: Analysis, Linear Algebra, Probability, Optimisation and OR, Graph Theory, Forecasting, Statistical Modelling, Econometric Theory